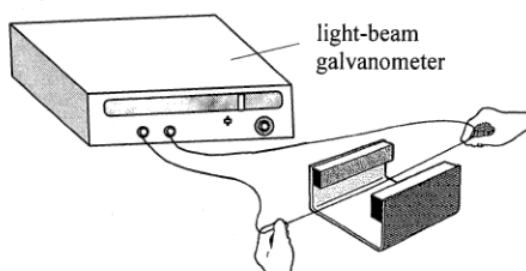


10.

Diagram for set-up:

1A



1A

1A

Connect the long wire to the galvanometer./Connect the apparatus as shown.
Put/Place/Move the wire across the magnetic field between the pair of magnets.

Relative movement: Move the wire vertically down across the field and then up across the field, the (light-spot of the) galvanometer would deflect to one side and then to the opposite side.

Polarities of magnet: Move the wire vertically down across the field, the (light-spot of the) galvanometer would deflect to one side and then reverse the polarities of the magnet to repeat the experiment, the galvanometer would deflect to the opposite side.

Direction of movement: Move the wire vertically up/down across the field, the (light-spot of the) galvanometer would have large deflection. No deflection when the wire is moved horizontally to the left/right in the field instead.

Orientation of conductor: With the wire perpendicular to the magnetic field, move it vertically up/down across the field, the (light-spot of the) galvanometer would have large deflection. No deflection when the wire is placed parallel to the field instead.

Rate of movement: Move the wire slowly across the field and then quickly across the field, the (light-spot of the) galvanometer would deflect more for a quicker movement or a faster rate.

No. of turns: Wind the wire into, say, a 10-turn coil and move it again across the field, the (light-spot of the) galvanometer would deflect more for more turns of

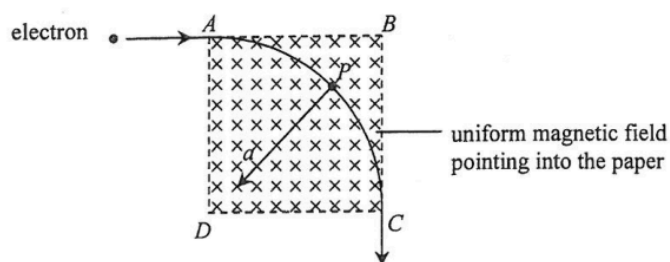
ANY
TWO
@2A

7

4. (a) (i) $F = qvB = (1.60 \times 10^{-19} \text{ C})(1.2 \times 10^7 \text{ m s}^{-1})(0.01 \text{ T})$
 $= 1.92 \times 10^{-14} \text{ N}$

1M
1A 2

(ii)

1A 1

4. (b) F is perpendicular to the electron's displacement / velocity,
Or electron only changes direction while speed remains unchanged,
no work is done, therefore k.e. remains unchanged.

1A

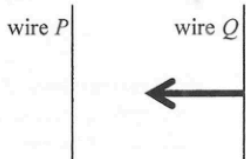
1A 2

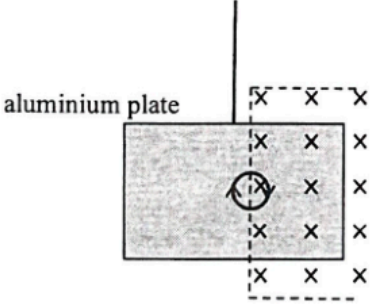
(c)
$$F = qvB = \frac{mv^2}{r}$$
$$\frac{v}{r} = \frac{qB}{m} = \text{constant}$$

1M

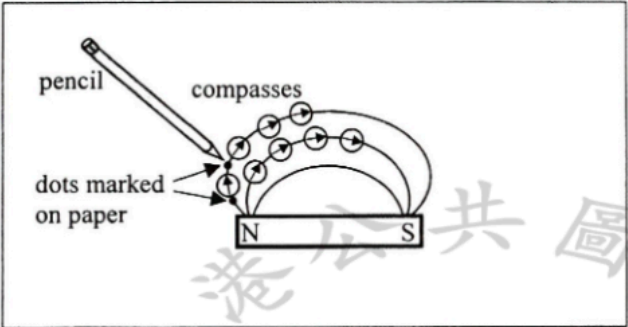
1A 2

v should be halved, i.e. $0.6 \times 10^7 \text{ m s}^{-1}$

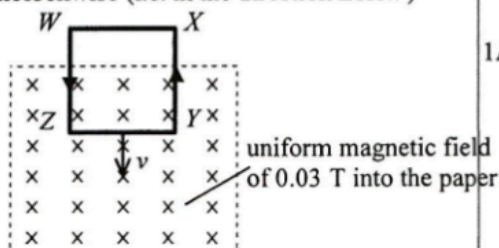
9.	(a)	(i)	The magnetic field at Q due to P points out of the paper.	1A	
				1	
	(ii)			1A	
				1	
	(iii)		The magnetic field at Q due to P is		
			$B_Q = \frac{\mu_0 I_P}{2\pi r}$	1M	
			For a certain length segment l of wire, the magnetic force is		
			$F = B_Q I_Q l \sin \theta$		
			$= \frac{\mu_0 I_P}{2\pi r} I_Q l$	1M	
			The magnetic force per unit length is		
			$F_l = \frac{F}{l} = \frac{\mu_0 I_P I_Q}{2\pi r}$	1M	
				3	
	(iv)		The two forces form an action and reaction pair, thus they are equal in magnitude.	1A	
				1A	
				2	
	(b)	(i)	As current passes in the same direction between neighbouring wire segments, the wire segments attract each other, and the solenoid is compressed.	1A	
				1A	
				2	
		(ii)	Current keeps on flowing in the same direction between neighbouring wire segments at each instant, thus the solenoid will be compressed due to magnetic force.	1A	
				1	

9. (a) (i)	By Lenz's law, an e.m.f. would be induced such that it opposes the change, i.e. decrease of magnetic flux (into the paper) by driving an induced current (clockwise) in the coil / circuit (complete).	1A 1A	
		2	
(ii)	$N\Delta\Phi = NBA$ $= (20)(0.3)(0.005)$ $= 0.03 \text{ Wb}$ $\xi = \frac{N\Delta\Phi}{\Delta t} = \frac{0.03}{0.5}$ $= 0.06 \text{ V (or 60 mV)}$	1A 1M 1A	
		3	
(b) (i)	The change of (magnetic) flux linkage is double that in (a)(ii), i.e. 0.06 Wb.	1M/1A	
		1	
	(ii) Direction of current : PQRS	1A	
		1	
(c) (i)		1A 1A	
		2	
	(ii) Move / swing to the right initially / momentarily / briefly.	1A	
		1	

3.	(a)	With superconductor / extremely low or nearly zero resistance, a much larger current flows in the coils producing a stronger magnetic field (with less heat loss due to the current flow).	1A	[MR: low R + large I – 2A marks large I only – 1 A mark low R only – 0 A mark]						
			1A							
			2	Poles						
				2N	2S/S+N	正	NA			
	(b)	P and Q : N (north pole) Repulsive force between like poles	1A	Reason	✓	2	0	1	1	
			1A							
			2	✗/NA	1	0	0	0		
	(c)	(i)	As the train is not in contact with the rail, <u>there is no friction / interaction between the train and the rail</u> , thus smoother due to less vibration.	1A	NA – Not answered					
	(ii)	There is no friction / interaction between the train and the rail , and the propulsion of the train only needs to work against air resistance, therefore much faster.	1A							
			2							

7. (a) (i) A: south pole (S) B: south pole (S)	1A	
	1	
(ii) (Towards) bottom / downwards	1A	
	1	
(iii) Iron filings come into physical contact with the magnet will never come off / are difficult to get rid of.	1A	
	1	
(b) (i) Successively place the compasses on the paper near one pole (say N) of the magnet such that the tip of each compass needle follows / lines up with the tail of the compass ahead.	1A	
Diagram	1A	
		
Denote the tip and tail (of each compass) with dots on the paper using the pencil <u>or</u> remove the compasses one by one and trace the (direction of) field line.	1A	
Draw / sketch a smooth curve representing a field line by joining up the series of dots/segments going from one pole to the other.	1A	
Repeat the above process by starting from different points around the magnet to get another/several field line(s).	1A	
	5	
(ii) Any ONE: - Compass is sensitive enough to explore very weak magnetic field such as the Earth's field. - The drawing can show the direction of the field lines / the drawing of the field will still be on the paper after removing the magnet. - It is difficult to draw individual field line separately in iron-filing method even if the filings are sprinkled very thinly and evenly.	1A	
	1	

9. (a) Induced current is anticlockwise (i.e. in the direction ZYXW)



1A

1

(b) Current $I = \frac{\varepsilon}{R} = \frac{Blv}{R}$ [Or $\varepsilon = \frac{N\Delta\Phi}{\Delta t} = \frac{B\Delta A}{\Delta t} = Blv = IR$]
 $0.01 = \frac{0.03(0.10)v}{4 \times 0.15}$
 $v = 2.0 \text{ m s}^{-1}$

1M

1A

Or

Power input $P = Fv = I^2 R$
 $(IlB)v = I^2 R$
 $Blv = IR$
 $(0.03)(0.10)v = (0.01)(4 \times 0.15)$
 $v = 2.0 \text{ m s}^{-1}$

1M

1A

2

(c) (i) $V_{YZ} = I(R_{ZW} + R_{WX} + R_{XY})$
 $= 0.01(0.15 + 0.15 + 0.15)$
 $= 0.0045 \text{ V} (= 4.5 \text{ mV})$

1M

1A

Or

$V_{YZ} = \varepsilon - IR_{YZ}$
 $= Blv - IR_{YZ}$
 $= 0.03(0.10)(2.0) - 0.01(0.15)$
 $= 0.0045 \text{ V}$

1M

1A

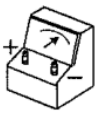
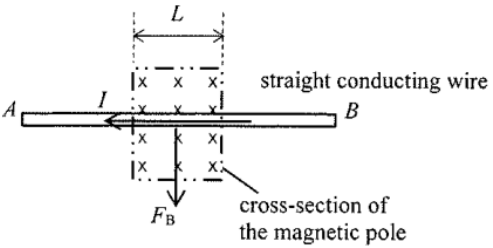
2

- (ii) As there is voltage drop (IR_{YZ}) within / across YZ,
 V_{YZ} is less than the induced e.m.f. across YZ ($\varepsilon = Blv$)

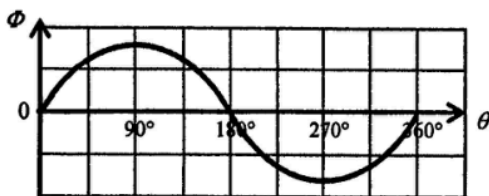
1A

Note:
 $\varepsilon - IR_{YZ} = V_{YZ}$
 $\varepsilon = 0.006 \text{ V}$

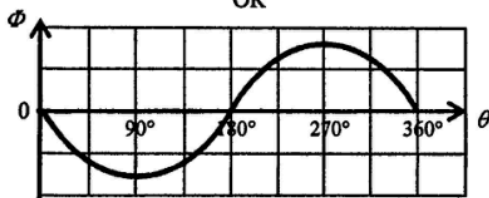
1

Solution		Marks	Remarks
8. (a)		1A	
		1	
(b)		3A	
		3	
(c)	magnetic force F_B		
	using a stronger magnet of the same dimensions	increase	1A
	using another support such that the position of the horizontal wire is slightly lowered	unchanged	1A
	using a longer wire and the current is kept unchanged	unchanged	1A
			3
(d) (i)	(Switch on the power source.) Adjust the power source to deliver a small current, say, 0.5 A to the wire and record the ammeter and the balance readings. Adjust the power source to increase the current stepwise (0.5 A) and record the corresponding balance reading.	1A	
		1A	
		2	
		1A	
		1	
(ii)	The balance reading increases linearly with the current.	1A	
		1	
(iii)	The graph does not pass through the origin because the balance has not been reset to zero initially before the experiment.	1A	
		1	Accept: because of the weight of the support / wire

8. (a)



OR



(b) (i) $\phi = BA \cos 0^\circ = 2.70 \times 10^{-4}$
 $B \times (9.00 \times 10^{-4}) \times 1 = 2.70 \times 10^{-4}$
 $B = 0.300 \text{ (T)}$

(ii) $\mathcal{E} = \left| -N \frac{\Delta \phi}{\Delta t} \right|$
 $= \left| -100 \times \frac{2.70 \times 10^{-4}}{0.01} \right|$
 $= 2.70 \text{ V}$

2A

2

1M

1A

2

1M

1A

2

Accept: -2.70 V